

Basics of Active Inference (Lecture) ~ Ben White ~

Active Inference for the Social Sciences 2023

Source: [YouTube](#) **Playlist:** Constructing cultural landscapes: Active Inference for the Social Sciences (2023 course) **Duration:** 1:32:55 | **Uploaded:** Unknown

Transcript method: auto_caption

all right hello and welcome everyone it is July 11 2023 we are in active inference for the social sciences and today is going to be a lecture by Ben White on basics of active inference the active inference agent Subban thank you for the lecture off to you and we're looking forward to it thank you very much yeah me too um very happy to be a part of this um to be kicking things off with the the first module on this course um I'm going to be talking about um the active inference agent and trying to cover some of the basics um I won't take too long introducing myself Ava or gave a very thorough introduction um in in his talk recently but I'm a second year PhD student in philosophy at the University of Sussex and I work with Andy Clark and with aval and some other people um using active inference to try and find things out about our relationship with technology and within the context of well-being and mental health the aims for today I'm going to try and provide quite a wide-ranging overview of how active inference has connected with areas of philosophical interest and particularly how they relate to the individual agent and the experience of the agent and how the agent finds themselves in the world so I've split this up um to look at several different defined topics so I'm going to move quite quickly through mind agency emotion and phenomenology and I'm going to say a little bit about the self as well and then there's going to be a case study at the end that I'm going to move through if we have time so one of the other aims that I have is to lay out very clearly some of the Core Concepts and core mechanisms like Precision weighting and prediction error generative models I'm not going to go into any technical details whatsoever this is all going to be fairly abstract my background and my interest in these Frameworks is philosophical so it's quite quite abstract in terms of how everything fits together and I'm going to try and bring some of these threads together to to build up a layered picture of what the active inference agent is like because this course is kind of aimed towards seeing how active influence supplies to larger scales Collective Behavior social norms socio-cultural Landscapes and I think in

order to do justice to those things we need to First have a good idea of the individual in active influence so we've got our work cut out for us um so I'll get started so before I dive in I wanted to just say a little bit about how these Frameworks hang together so um active inferences based on call for instance free energy principle I think most of us are familiar with that but it's the free energy principle just states that in order to persist Through Time biological organisms must occupy only those states that it would expect to occupy given the type of thing that it is so free energy is essentially a measure of the disattunement between a system and its environment and I'm going to say say a little bit more about a tune and how that gets cashed out in various ways as we go on active inference is a process theory that essentially explains how embodied organisms actually go about remaining in those expected States so how we actually go about minimizing free energy and the idea is that all adaptive behavior is explained by agents garnering evidence to confirm um their own expectations the predictive processing for the purposes of the lecture today I'm going to treat these as synonymous because most of the work that I kind of came up through as I was learning about this was predictive processing and all of the papers um that we're using today that refer to predictive processing refer to a predictive processing that I take to be more or less synonymous with active inference so that's a a very embodied inactive flavor of predictive processing I think generally the difference is that predictive processing can be a much broader term and predictive processing can also apply to passive models of perception but we're not going to concern ourselves with that okay so some key concepts of active inference slash predictive processing then so the framework says that agents embody or agents have um we won't concern ourselves with the with the Nuance there but they have a generative model which is essentially um a kind of understanding or a model of the regularities that underpin the Dynamics between the agent and the environment so another way to think about this in very rough terms is a kind of mental model of the regularities in the world and how the agent exists in the world and using that model the agent can generate predictions about its own sensory States so given the model what kinds of sensory States would it expect to find itself in in a particular set of circumstances and so where those predictions don't actually match the sensory inputs so where there's a discrepancy between the actual income incense data and the prediction prediction errors are generated and prediction errors are going to flow upwards through the hierarchy so the um the generative model is said to be a hierarchical model and predictions flow downwards some prediction errors flow upwards and the sole imperative of the brain body system according to this framework is to minimize those prediction errors so a good way to kind of get a little

handle on this is to think about just the example of perception so perception traditionally so I'm talking about visual perception here it's traditionally been understood as a bottom-up process and what that means is that the brain waits for incoming sensory signals and then it processes them as they come in and it combines them in kind of increasingly abstract ways and what we see is some combination of those sensory signals that have come in and been processed and combined so if you think about me leaving my uh apartment in the morning and stepping out onto the street and seeing some cars what's happening there is some sensory information is hitting my retina and it's being processed at first with for very basic features like light shade edges and so on and so forth and then it's being combined with my understanding of what car is or what a bus stop is and that gives me my kind of visual field predictive processing aimlessly at this point flips this picture upside down on its head so visual perception on the predictive process and is a top-down Affair so the generative model is going to encode multi-level expectations about the likely scene so just to be very clear when we talk about multi-level and hierarchical the higher up is the higher up levels are going to be tracking longer time scales and higher levels of abstraction and the lower levels are going to be much faster and much more concrete so a changeable scene so there's going to be some variation in what I see when I leave my apartment every day is going to mean that there's always going to be some error in that picture so it's never going to be the same two days in a row or it's extremely unlikely to be so no matter how well I can predict what I'm going to see when I leave my house it's always going to be cleaned up by some prediction errors okay so essentially when I step outside and I step into the street my brain is already anticipating what it's going to see out there okay so I know where particular cars are going to be parked I know where the bus stop is so on and so forth and the perceptual experience that we have is a construction it's uh we actually experience the expectation that has been kind of cleaned up post-prediction error and there's a little bit of debate there about exactly what it is that we experience but um for our purposes today um we can say that visual perception is essentially a construction so on the face of it prediction errors can be minimized using one of two strategies and this is going to be really really important so we have what I just described there which is perceptual inference that's where we revise our predictions to better fit the evidence so essentially we update our model to better reflect the regularities in the environment or we can engage in active inference which is where the framework gets its name from which is where we essentially make the evidence fit the model okay so that's where we in some sense update the world so it conforms to the original prediction and active inference agents bring about

those expected states by sampling the world in a biased way so they have certain expectations and then we sample the world to confirm the predictions and I put perceptual inference there in scare quotes because perceptual inference on um kind of readings of most readings of active inference just becomes a kind of action Readiness or as ramstead and colleagues but a state estimation so it's just perception and action are just wrapped up in this continuous loop but it's really all about the action it's all really all about bringing about those expected States and perception is just kind of in service of that process and so bruenberg and colleagues refer to the active inference agent as a crooked scientist which is uh something I really like I think that's a very Charming image so we are scientists that are engaging in some very dodgy Behavior because we're not really interested in uh we're not really interested in having an accurate model we're more interested in um confirming our original predictions Okay so that was a very very rapid overview of some of the Core Concepts and mechanisms but they're going to crop up continuously so I think we're going to get um a much better grasp on them as we go on in this section I'm going to start to expand on some of those things that we've just put on the table and I'm going to begin to flesh them out in terms of thinking about how active inference agents actually exist in the world so I should say I put this here to remind me so those of you who are taking part in the course um I'm going to make available a document with all of the readings that I firstly all of the readings that I've used to make these slides and then a bunch of other readings as well that I think can be very relevant and um obviously they're going to be organized in terms of the structure of this presentation as well so this is just I put a little sample up on these slides but there's going to be much more in the document so I think it's always good to start with some guiding questions so I've thrown up three here the first is how does active inference characterize cognition so we've just looked at um predictions and prediction errors but we're going to expand on that a little bit and we're going to ask what kinds of processes does cognition involve how is cognition related to the material environment this is going to be something um how cognition relates to the environment that's going to be very important in subsequent weeks and we're going to ask quite briefly what kinds of vehicles might realize cognitive processing okay so it is here's a couple of whales whales are going to be a recurring theme in this lecture um but I wanted to take a step back and ask about Attunement because this is one of those Concepts that gets thrown around a lot and I think once it's grasped it becomes very very clear and very easy to use but I think prior to it being grasped it can be a little bit opaque and a bit slippery so on the left we have a humpback whale that was found dead in the Amazon rainforest allegedly I don't know if this was true and on the

right we have a whale in its natural environment and I think clearly the intuition here is that in one case the whale is very well attuned to its environment and in the other case it's not and I think that we can start by asking what is it about the ocean that allows the whale to be well attuned and what I want to say here is that the particular Dynamics Between the Ocean and the whale allow the whale to do certain things that it can't do when it's in the Amazon so in other words we might say that the ocean affords certain things to the whale specifically filter feeding um being able to move its vast weight around with ease um things that unfortunately are wailing that Amazon can't do so affordances are opportunities for Action um they they are things roughly speaking in the environment that the agent can act upon and there's some debate about the proper way to really characterize affordances but I like to think of them as relational properties emerging from an environmental feature and some skilled embodied ability of the agent so the same thing in the environment will not afford the same thing to different organisms so for example uh soil to a human agent might afford walking on or laying down on but to an earthworm it affords very different things and also even amongst the same organisms with humans especially different surfaces and features of the environment will afford different things depending on your particular skill set um and agents are going to go about minimizing their uncertainty so minimizing prediction errors staying attuned to their environment through an engagement with affordances and this is going to be something that is going to drift away slightly as we move through some sections and then it's going to come back very heavily at the end and I should say that the idea of affordances to my knowledge it first emerges in the work of James Gibson the ecological approach to visual perception I think is chapter eight in that book that really lays out a theory of performances so this idea that systems or organisms maintain their organization through ongoing action um means that active inference has been labeled as quintessentially inactive by Corporation and I think that this is a nice in inactivism is an older framework from theoretical cognitive science and I think it's a nice way to understand some of the key features of active inference but it does come with a little bit of historical philosophical baggage and I'm going to try and unpack that really quickly um because I want to unpack exactly what it is that we're committing ourselves to when we think about cognition in in activist terms so inactivism comes in various strands very very quickly and roughly you have Auto poetic inactivism which commits itself to what's known as the Mind life continuity thesis this is essentially that the structures that um give a ground to life itself are the same structures that instantiate cognitive processes um sensing on this view is teleological because it's Oriental oriented around supporting vital systems so things in the

environment have intrinsic meaning to the organism because they support the vital um functioning of the organism in some way sensory motor in activism emphasizes how mobile and embodied agents essentially enact the grounds and conditions for their own sensory engagements with the world so the world is not a kind of neutral static given that we're parachuted into actually the conditions through which we sense the world are the conditions that we um we can move and kind of interact with the world in various ways um and then there's radical inactivism and this is um The View that cognition is purely grounded in these agent environment Dynamics it's a purely dynamicist account of cognition and there's no role whatsoever for internal representations so certainly in certain brands of sensory motor in activism there remains a role for internal manipulation of representations and mental representations are just mental states that are about something in the world so some mental state is reconstructing some feature of the world and this all harks back to um kind of old debates in philosophy of cognitive science between internalist Notions of cognition and more embodied embedded extended and will take a more of a look at that in a second but the question here is what are we committing ourselves to with active inference active inference and I need to be a little bit careful because I think that people are going to disagree with this to a certain extents but I think we're certainly committing ourselves to sensory motor in activism I think that's fairly uncontroversial and I would say as well that the the underpinning of active inference by the free energy principle by my lights also commits us or it certainly Chimes very deeply with the core tenets of Auto poetic inactivism but I think what's really interesting in trying to understand cognition in the active inference agent is to understand what role if any there are for representations if you have any familiarity with the literature on active inference you'll know that active inference people are always banging on about models um various other computationally sounding things and so the worry here is at least for um well it's not necessarily A worry but the thought the thought would be are we committing ourselves to a form of cognitivism and again cognitivism is this is generally considered to be a old-fashioned notion of cognition as like computation over symbolic representations in the brain According to some syntactic set of rules or and typically over the last couple of decades in cognitive science people have been moving away from that view so it would be interesting if we were committing ourselves to something that we'd already moved away from and some people think that we we are making that commitment so Jakob Howie for example has argued in various places um that essentially predictive processing or active inference commits us to these very rich reconstructive Notions of representation and even Andy Clark um Jacob Howie's Boyle in

many many places has said himself that he thinks that representations play a role here as well so in a talk in 2016 Clark said it's internal representation I think it is these are representation using stories but it's not internal representation in quite the way that we originally thought about it so there's some differences perhaps major differences between Howie and Clark there but others have pushed back so there's a paper by Maxwell ramstead and colleagues from 2020 I think it's a tale of two densities paper where they say that a proper understanding of the generative models in active inference suggests that there's no role for um so this is the kind of anti-howie view I really like this paper from Axel constant and Andy Clark and Carl friston which essentially tries to broker some kind of piece in a debate that in some form or another has been raging for decades so it'd be quite nice to finally put this to bed um so they argue that we can find in the formalisms of active inference we can find grounds for an Armistice here um so they argue that the formalisms of active inference show that quote representational and non-representational cognitive processes can be implemented by the brain under active inference and they do this by showing that active inference accommodates a dual architecture so they show that on the one hand um in some sense the brain has to represent because under active inference uh action selection policies are happening over extended time scales and so the agent has to engage in some kind of counterfactual thinking so if I do x what happens if I do y what happens and so on and the argument very roughly speaking is that that is going to entail some kind of representational some use of representations but on the other hand they argue that what they call deontic actions are processed through different mappings in the generative model and they're directly triggered by sensory input so a deontic action is an action that is ingrained by our sense of the expectations of others so the example that the authors use is stopping at some red traffic lights in the middle of the night when there's nobody else around so in a sense the the claim is that we through a kind of social conditioning we know what the what other people would expect us to do in those circumstances and that allows for this kind of dynamic reflective reflexive um action policies I think and in those cases um there's no room for representer or there's no need for representations there um I would uh definitely suggest people interested in this go and read this paper because you might already have thought you might already have thought of some potential objections here uh to the um the claim that representations aren't needed for the deontic action and they do go into some detail in talking about uh potential objections there that's the kind of status of the role of um it's a live debate not everybody agrees but it certainly looks as though active inference has the formal tools to accommodate both sides of the debate and it might turn out that that's a

good thing because we can make everybody happy or it might turn out to be a bad thing because you might think well we've not really settled anything one way or another I wanted to take some time though um to mention um Extended active inference so I've been talking about inactivism and I've mentioned that uh I've mentioned this debate between cognitivism and older older ways of thinking about sorry cognitivism and newer ways of thinking about cognition as embodied or embedded and these are part of what's known as the 4E Paradigm and clearly I think we are talking about inactive cognition and you get embodied and embedded kind of for free with this so the only one left on the table is the extended mind and this was just the claim that under some conditions um the system that's um realizing cognitive processing can literally extend outwards from the mind so I'll put some nice readings for this up if anybody's unfamiliar and interested but it's usually the most controversial of the four is and Andy Clark who is one of the original authors of the extended mind paper really thinks that uh active influence accommodates uh the extended mind Theory and it's not just that it accommodates it but it actually solves some old problems in the extended mind about knowing how and why the brain uh knows when and where to recruit certain extraneural resources and part of the argument for this what really sits at the core is um Clark develops this argument based on the fact that the generative models in active inference are always in this always making a trade-off between accuracy and complexity so you want your model of the world to be accurate enough that you can act effectively on the world but you also want to minimize the metabolic costs of having models that are overly complex and so Clark utilizes this insight to to make the case that this explains that in some cases we as agents we make use of these um extra neural kind of props and tools and things um and he says that the the imperative to reduce uncertainty provides a location neutral cost function that's always been missing from the older arguments around extended mind so the upshot to this I think is that you end up with a view of Agents as deeply coupled with their environments so under active inference it's just nonsensical to try and think about agents as decoupled from their environment the generative models themselves are of the Dynamics between the agent and the environment so these two things can't be decoupled um perception action Loops so remember I said that perceptual and active inference are essentially the same thing they're the part of this circular causality and they see the agent constantly producing their own worlds they occupy the states that they're expecting to through action and we might think based on the literature that I've gone through that there's some role here for representations so we're not committing ourselves um the active inference agent is not one that fits with very radical interpretations of inactivism um and inact

active inference agents are porous they're open to recruiting external material features of the environment to reduce complexity and therefore minimize the metabolic costs of realizing those expected States so that was a bit of um active inference set against the backdrop of some older questions in philosophy of mind theoretical cognitive science but I'm going to jump straight into agency and motivation now because there's a lot to get through here um I'm going to start with a dark room problem this is a canonical philosophical worry in active inference so a lot of people I think will be familiar with this and in really brief terms it's really easy to State um The Challenge here is to say if it's all about minimizing prediction error why don't we just find the most predictable environments possible and stay there and live a nice error-free life so Andy Clark paints a picture of a dark room and you can just be strapped up to a to a intravenous drip feeding you the nutrients you need and you can just be really happy but clearly we don't do that um active inference or sorry agents in the world us we uh generally typically sometimes we do enjoy dark rooms but most of the time we don't for any extended period of time and so there's a challenge here about how active inference can account for what appear to be motivated agential actions that necessarily mean we're confronted with some degree of uncertainty um and I'm going to set active inference up in a kind of dialectic with folk psychology here because folk psychology faces No Such concerns um so to be very clear folk psychology is just the kind of propositional linguistic framework that the folk use to talk about and describe motivated action so it's a language we typically use every day to describe the reasons people are doing things so uh folk psychology has this belief desire intention schema where if I want to explain my friend um going to grab a beer I would say my friend believes there's beer in the fridge my friend desires beer and so those are combined to form the intention to go and get beer so folk psychology doesn't have these kinds of problems um so just a little bit more on the dark room problem and another whale um so the dark room problem invites more than one answer we're going to see more than one answer to it as we go along um one response is to talk about what drives curiosity and play and exploration and we're gonna we are gonna see more of that but the First Response offered by Friston Thornton and Clark is to highlight the fact that organisms have basic needs to eat drink and reproduce what you might call evolutionarily ingrained needs and that those needs are going to necessitate some form of exploratory Behavior so staying in the dark room will inevitably under most circumstances going to lead to physical dissipation so the question is are these kinds of appeals enough and I wanted to pull up a paper by Colin Klein 2018 that I think is like a really nice articulation of these challenges ultimately I think he's wrong and it falls flat

but I think it's a strong articulation so Klein argues that predictions alone simply can't account for the motivational character of these needs so Klein says look it's fine that you've got these needs nobody's going to deny that but you just have this one primitive to describe how those needs motivate you and prediction is just not going to do it so prediction alone as your sole primitive is not going to be able it's like predicting is not the same thing as being motivated by something um and Klein is really not happy with appeals to these phenotypic needs and he construes these in the relevance of The evolutionary history of the organism and goes into a little bit of detail so he says look you're right these expectations are either going to be about States possible States so the state of being fed or clothed or warm or they're going to be about State transitions about what the appropriate action is in one state to move to another state and he says expectations about States themselves aren't going to work because you have this problem of accounting for novelty so it might be the case that in my evolutionary past all of my ancestors have had the same expectations about which states they're going to occupy but there are lots of states in my life that I could be motivated to occupy that there's simply no evolutionary history to account for and he says there's a very similar or a similarly difficult problem with State transitions where essentially what we're talking about with State transitions the most appropriate actions so I have expectations about the actions that are going to fill for that I need to take under certain circumstances that are going to get me to a particular expected State and uh Klein's worry is that there's just going to be too many possible actions in any given situation so there's just a kind of um a kind of explosion that's just going to be unaccountable for there's just going to be too many different ways of getting to an expected State um that won't be uh we won't be able to account for in our past so it's fairly robust thorough challenge you've built on the dark room problem and I think actually there are it's a worry with two faces so I think um the first is about how active inference can describe agential Behavior with its ostia purely doxastic landscape so if you imagine um somebody looking at their house burning down it seems like you need more than beliefs to explain any course of action they take so there has to be desire in there somewhere so this is an example Andy Clark uses in one of his papers and he says you know um the desire to run in and rescue some kind of some item that you want from the house or the desire to claim on the insurance is the thing that's going to dictate how you explain the person's Behavior so that's the first problem we need to look at how folk psychology uh tessellates with uh prediction and the other problem is maybe a little bit maybe a little bit deeper and that's to ask how does active inference actually account for agential Behavior so can we even tell some story using just predictions and

prediction error about what really moves agents to act so firstly thinking about folk psychology there's a couple of ways to go here and I think ultimately we can answer this worry so Ryan Smith and colleagues in 2022 um in their paper active inference models do not contradict folk psychology they argue that the formalisms of active inference can accommodate a belief desire distinction now I'll admit the that's a very math heavy paper and it's a little bit beyond my capabilities but the claim here is simply that there's no contradiction that actually the worry that the important thing we have in folk psychology is desire and we're not getting that in active inference this is just an unfounded worry but there's another strategy as well so Joe Dewhurst in his 2017 paper he employs a different strategy and he says look what are we even doing with folk psychology let's not kind of be too hasty um and we we need to think about what are we doing when we talk about these uh these propositional kind of States um so he says the first option is to say well we're just being realists here so when I say that X desires y I'm actually positing the real existence of a fine-grained mental state in ex's head and he thinks that this is actually the wrong way to think about things he thinks that folk psychology actually it's an instrumentalist tool it's a kind of framework that we use just for picking out course grained behavioral patterns in some useful way so if we agree with Dewhurst here and we just become instrumentalists about folk psychology again it looks as though there's different um we don't we're not forced to kind of worry that um there's some kind of empirical problem here where the mechanisms of active inference are in some sense incommensurable with real properties that we're positing the existence of because we're just going to say that those things don't really exist in the way that we've typically assumed that that they might um so either way um whichever route you go down here it looks like active inference agents have beliefs and that we can we we're going to be fine in describing motivated action so what about the second problem so Andy Clark in 2020 authored a direct response decline um so Clark wants to argue that actually using predictions and prediction errors and uh Precision waiting which I'll explain in a second um we can get a picture of agential so the first thing Clark does is to outline the fact that under active inference we have a very nice well-worn account of embodied action so in other words what it takes to actually move the body to perform different actions is perfectly folded into active inference under the same kind of computational flow of prediction error minimization just in a nutshell for me to move my arm what I'm doing is making a prediction that my arm is in some place and then minimizing the prediction error um so once that's on the table Clark says okay that's fine uh we have this account of action but that's not really what Klein means um so the move clock

makes is to appeal to the hierarchical nested structure of the generative model and to say that higher level beliefs which are temporal about temporally extended future States these are going to perform a really important role as a kind of controller for lower level actions so the idea is that if I have um a high level temporally extended expectation to attain a PhD what happens is that high level belief becomes unpacked at lower levels in the hierarchy and what you get are more fine-grained faster action policies that the system expects to help support that higher level expectation there's a mechanism in play here called Precision weighting which applies to um prediction error signals because obviously um not all of the incoming not all of the predictions we make about the environment are going to be equally important we're not going to have equal confidence in all of our actions to bring about an expected State and so what Precision weighting does is essentially communicate how much confidence we have in a particular action policy to move us into that expected State under perceptual inference and precision weighting just signals how newsworthy prediction errors are so how important they are for the system so Precision weighting drives attention for example um when we're talking about predictive processing but in this case Clark's claim is that when you have this kind of nested hierarchy of expectations so you have temporally extended expectations at the high level unpacked into a kind of web of lower level action policies and precision waiting is kind of modulating the system's expectations about which of those policies are most likely to bring about the expected state you get something that looks a lot like motivated directed action and agency is kind of inherent in this model because the agent has to the system rather has to model itself as a cause of those expected States so if I if I have these high level expectations I in a very inherent way have to model my own cause or efficacy to bring about those States so on a very low level the system has to model some form of agent because I can be the cause of change to my own sensory States so I can cover my own eyes and so on and so forth so kind of on this view agency is a latent variable in these systems when it's making these predictions so it just kind of turns up and clock argues that you you essentially end up with something that looks very much like um agents on the folk psychology but he thinks this is a better view because we end up with a quite different and arguably much more unified internal architecture trading in a single currency so Clark thinks look okay Klein's worry is we have this one primitive we have just prediction and the concern is that we're not going to be able to explain agential action but actually there are ways that we can appeal to the hierarchical structure of the model and precision waiting and it turns out that this is a much more elegant picture as well but Klein might press the worry and anybody listening might kind of think well we've

kind of done some work there but why do we expect the things that we expect in the first place [Applause] so why is my long-term expectation to attain a PhD and not to join the French Foreign Legion for example okay what means that I expect one thing rather than another and here Clark just bites a bullet and says nobody can explain that what we're kind of we've kind of shifted to talking about questions about you know spookier questions about Free Will and Clark says look if this is what you want if this is what you're demanding then you have to recognize that simply appealing to folk psychology doesn't get you that either okay so Klein or you know people um with these kinds of worries need to be careful in how much they demand from the from the active inference agent and I think the real answer to that question actually this is more speculative but the real answer to that a deeper answer to these kinds of questions will lie in the kinds of things that we're going to look at in subsequent weeks so for example the reason I want I have the expectation to attain a PhD rather than join the French Foreign Legion is because I exist in a particular kind of socio-cultural landscape with specific norms and a lot of my actions and beliefs are tempered and shaped by the expectations of people around me so this is just an overview we went over the dark room problem um why do we do anything at all if it's all about uncertain if it's all about certainty and prediction error minimization this worry I said you can kind of um highlight two aspects of this worry one is an actual account of agency under active inference which we have we have this Andy Clark model of agency to do with hierarchical nested unfolding action policies over time and different precision mappings and then we have this worry about the relationship with folk psychology so how do we describe um agential Behavior motivated Behavior how do we describe the behavior of Agents if we only have recourse to uh and as we saw there's um one argument here is that actually active inference the formalisms can accommodate this belief desire distinction and the other move that I really like is to just say look folk psychology is just this instrumental tool it is in effect a kind of it's part of the socio-cultural landscape that we use to shape um each other's Expectations by this putting these labels to explain our actions it's a kind of form of active influence itself so there's this kind of philosophically interesting debate about folk psychology um but we it turns out that everything's okay and active inference agents are in fact agents with beliefs desires and hopes and dreams and so on that's agency moving we're making good time but okay so in this section I'm going to introduce quite a lot of stuff um but it's going to be all tied together hopefully rather neatly in the case study at the end I'm going to talk about allostasis and emotion I'm going to say very very briefly a little bit about the self as well uh here's some examples of some of the core readings I've used again

these are all going to be up in the document as well which will be made available um and I thoroughly advise you to just have a browse through these there's some really interesting stuff there um I'm not going to talk about um psychedelics or meditation or depersonalization but there is some really really cool active inference literature on those topics as well particularly papers on psychedelics and ego dissolution by George Dean but new paper just came out by Jonas Mago and Robert Carhartt Harris and I think Fernando Rosas as well I think I might be mistaken on the authors there but it's a very new paper uh okay so the the thing that is going to underpin a lot of what I'm now going to go on to say is going to be this notion of allostasis and this plays a major role um in a lot of work on emotion and the self in active inference Lisa Feldman Barrett whose theory of emotion we're going to look at in a second has argued that the brain's fundamental purpose what it evolved to do through that imperative to minimize prediction error is to manage the body's metabolic budget so the first prior is uh kind of our most expected State um the first prior in the sense of like our strongest prior um prior probability our highest expectation is to have um good allostatic control and I'm getting ahead of myself there all that means is that it's we're balancing our body's metabolic budgets or like energy budget in a in an effective way error signals like I said before that are highly precise are those that are weighted very highly so prediction error signals that have high Precision waiting are the synaptic gain on those error signals is going to be increased and that's going to drive learning which means in future our expectations around those kinds of error signals is going to be we're going to pay more attention to those um and the most highly weighted error signals that we're going to get are to do with our own homeostatic set points within our own body okay so if my uh temperature started to rise rapidly my body temperature started to rise or drop rapidly um that prediction error signal those the violation of those expectations is really going to grab my attention and I'm going to do something about it okay and one way that we maintain these homeostatic States is through the body's own autonomic reactions so where I to start getting very cold um I would probably start to shiver and so the body is trying to kind of Rectify those deviations from homeostatic set points but another thing that um organisms engage in and humans do this to really remarkable degrees and again this is going to be something that I suspect is going to come out in the subsequent weeks is that we engage in uh quite complex and sophisticated anticipatory actions to maintain homeostasis and this is something called allostasis so allostatic control which I mentioned earlier is this first prior is essentially the regulation of homeostasis through action so it's actions that I can take to ensure that I stay within those important bounds so shivering is good but living in a house with

central heating is better having a warm bed and living in a community where we've had these people of kind of Engineers have worked to create a kind of material environment that acts as a kind of background support to a lot of these needs um something that's another important concept here is allostatic load so this is essentially the result of not having good allostatic control so again allostatic control is just engaging effectively in actions that are maintaining those set points whereas allostatic load is the physiological costs to the agent of losing that control so there's so many studies here that have just shown the immense physiological costs of operating Beyond those set points and not having good control so allostatic load has been linked with reduced cardiac health sleep problems um uh gut issues mental health issues depression uh basically everything it's really really really bad and so good allostatic control is very important and as we're going to see it underpins quite a few very important things really like we could dedicate a whole hour to looking at emotion but I'm going to blast through it very very quickly right now and this slide I should say one of the core readings on the document and the reading that I'm basing this on is Lisa Feldman Barrett's 2017 paper the theory of constructed emotion the main idea here is this is a this is one of those theories that it on the face of it it seems quite simple but actually there's quite a lot of subtlety and Nuance to it um allostatic control means inferring the causes of changes to the body's internal state so in other words if I want to keep good allostatic control I need to know what's not just what's going on out there but I need to know what's going on inside me as well okay so if there's some crucial change happening I need to be on top of it and this happens through exactly the same process as the same predictive error minimization process that I'm engaging with with the outside world so the brain is making predictions about its own internal States and then it's going to engage in some kind of action to minimize any prediction errors and these prediction errors are going to be very highly weighted they're very important but here's an interesting thing so there was a very unethical experiment done by uh shakhter and singer I think in the 60s who they were emotion researchers and they conducted an experiment where they uh injected uh subjects without the subject's knowledge I think um they injected them with adrenaline and then they exposed the subjects to different uh stimuli so some subjects were exposed to one type of stimulus and some subjects were exposed to another and even though the subjects had essentially been um they'd been pushed to have the same to experience the same physiological changes in their body they reported um different emotional like different kinds of feelings depended on depending on the stimulus that they'd been exposed to so this highlights something really important and again I feel compelled to say that I am moving

through this incredibly quickly there's so much to say here but the the nub of it is this in order to keep good allostatic control in order to know what to do the brain needs to contextualize those changes in the body because there's not a one-to-one mapping between changes in the body and an external stimulus okay so sometimes changes physiological changes in our body are about something in the world they're a response to something um and it's very very important that the brain is kind of taking a Confluence of those inputs to try and contextualize okay what's actually happening and what do I need to do so an example of this is uh if you imagine running into a bear while you're hiking in Wyoming which is something that's actually happened to me I've experienced it it's terrifying and you imagine going on a scary or a date that you're really nervous about the physiological changes that are happening in the body are give or take more or less the same so perspiration elevated heart rate breathing changes um so the underlying physiological footprint is very very very similar but what you need to do to manage that situation that apt action is very very different and so the claim here the kind of the kind of revolutionary claim in a way is that emotions the things we experience subjectively are constructions just like percepts are so just like my experience of the car when I step outside my flat is a construction that explains away the prediction error and gives me a cleaned up picture of the world emotions are essentially doing the same thing they're explaining away these prediction errors and telling me this is the best explanation of these changes within your body contextualized against this external stimulus so in the case of the date apt action is to like chill out and be normal right to a certain extent and in the case of the bear it's you get the hell out of that although interestingly you don't running away from a bear is a terrible idea um you should not turn your back and run away but you get the idea that it's going to be what you should do in each situation is completely different so that's emotion for the active inference agent it's and something I don't want to I don't want to lose here is what you get with this is really kind of cool because it turns out emotions and subjects subject we're kind of moving into subjectivity here it turns out emotions are really they're just part of the same cognitive architecture so there's no difference here there's no dualism between his thinking and here's emotion and emotion is this kind of fuzzy stuff uh that we're not really interested in actually emotion subjective embodied feeling is core it's absolutely vital to understanding how we maintain this allostatic control and how we engage in adaptive behaviors and how we remain gripped to our affordances in the environment and the notion of grip is going to be something that comes up again very soon so very very briefly because this is something that I've really already talked about but I want to put a slightly different spin on it right now I've already said

the allostatic control for humans can involve very very temporally extended time scales and we saw this with Andy's model of uh agency so if you think about something like saving into a retirement fund for example okay we're doing something that the idea is that maybe decades from now is gonna kind of uh minor spikes in prediction error called the disappointment of having not having as much money right now is going to pay off later these are the kind of things that that humans engage in and our generative models have to be temporally thick so that George Dean writes that the levels of temporal extension are hierarchically interlocked so shorter time scales inform expectations of what the agent can do over longer time scales so these different levels of the hierarchy are in a kind of dialogue with each other telling you what kinds of things are possible and what kinds of things you should expect and when we were talking about agency we talked about the way that agencies kind of baked into active inference because in order for any of this to work the system has to have as a hypothesis I can do things okay um and this is uh more or less taken to be the same um the same thing that underpins the sense of self under active inference so the system implicitly refer infers its own ability to bring about intended sensory consequences and so Dean writes here quote in Princeton it is in this sense that implicit in a model of sampling is a representation or sense of agency a minimal sense of self it's kind of like the flip side of agency comes baked in here so one has to model oneself as a potential cause of sensory inputs so George Dean and colleagues I think George Dean Mark Miller and Sam Wilkinson propose a model of the self and the active inference called the allostatic control model um and the move they make here is that we have to recognize that it's not just our self-efficacy that has to be modeled we also have to have some kind of idea in there of how we're actually doing so we have to model our own capabilities in order to set Precision waiting on future um so we don't simply infer that we can be the cause of sensory States but also infer our own capabilities to successfully bring about expected States at longer time scales so the the phrasing that they use is tracking the performance or Fitness of the model over part of that tracking is going to be affective and emotional and so there are strong links we're kind of really digging our hooks into subjectivity at this point but the the claim with the allostatic control model is you get this rather than just a minimal self you actually get a longer term phenomenological self emerge from these interlocking time scales which are modeling not only the system the agent's own causal efficacy but also modeling how good your own model is at doing what you need it to do and like I said I said when I was talking about the reading I and I want to flag this again um because I've really really not doing Justice um to this in the way that I should but this has

formed the basis for a lot of really interesting work on phenomena like depersonalization [Music] um on the role of psychedelics and the potential for psychedelics in therapy and in meditation ego dissolution and kind of um that kind of stuff in in meditative settings so anybody with any interest in the self or in those kinds of practices should definitely go and give a closer look to these readings stood up I'll make sure in the document okay so quick overview here so active inference agents maintain homeostasis through anticipatory action and this is something we call allostatic control to do this the system must engage in interceptive prediction over its own internal States and I actually I kind of I think I've got time to do this so I'll just deviate slightly um the topic of interceptive inference is really interesting in the context of active inference so I I want to flag work by Sarah Garfinkel uh who I believe is at UCL who used to be at Sussex and um Professor Garfinkel's work looks at um the the central importance of uh interceptive predictions in various psychopathologies especially I think she's looked very closely at PTSD and anxiety disorders and it turns out that an agent's accuracy in their introspective predictions is a good indicator as to the kinds of symptoms that they're gonna like the strength and severity of their symptoms in these kinds of disorders um and what she's shown is that if you can increase the accuracy of interceptive predictions this forms the basis of treatment for these symptoms so in short the better you are at interceptive predicting I think one of the measures that they use there in her lab is how well people can track their own heartbeats without feeling their pulse so if you just sit there and try and say like do do my heart's beating like this then the more accurate you are and the more any training you can be given to be more accurate it forms the basis of effective treatments for these kinds of symptoms um so I just wanted to flag that because the world of research on in in interceptive inferences uh large diverse vibrant super interesting and very much worth our time um okay where was I here okay so to to maintain other static control the system has to engage in interceptive prediction over its internal States and then to best explain changes in those internal States the system has to contextualize those changes against some extra receptive uh input and that's emotion categories serve that role so just to be very clear on this view this emotions are constructions um there are some important deviations from other theories of emotion here um particularly what uh Barrett calls the basic theory of emotion um so on her view emotions themselves have no underlying kind of essence or neural footprints and there's a ongoing live debate there as well between these theories and also just at the end the good out of static control in human agents is this very temporally thick uh phenomena so we engage in very long-term planning that involves a lot of the kinds of counterfactual representations that

we talked about with the actual constant paper and this has hypothesized to form the basis of our phenomenological sense of self okay we're up to an hour um good we're running on time uh this is the final section and um it's probably going to be probably going to be the longest section what I want to do here is I want to introduce I want to try and pull together some of the things I've been talking about because it's been very fast and very abstract and I want to try and pull some of these things together in a slightly more concrete example so the case study I'm going to use is addiction so there's a particular account of addiction in the predictive process in literature that I think in a very succinct Sharp Way highlights a lot of the things I've been talking about and puts them within a context that is quite Stark and easy to understand um but in order to do that I'm going to have to introduce a new um slightly more difficult to understand mechanism within the framework that crops up in a lot of the predictive process in literature and what this is going to do hopefully as well is give us some a direct bridge to phenomenology so when we talk about things like neuro computational phenomenology um this the things I'm going to be talking about here have underpinned a lot of the a lot of the work in that area so these are some of the readings we're going to look at some work by uh mostly um a lot of the papers we're going to look at uh first author Mark Miller or Mark Miller's been the second or third author and I bring that up because in the discussion section um on the 25th when we have an hour to kind of unpack and answer people's questions we're actually going to be joined by Mark Miller so if anybody has any specific questions about the things I'm going to talk about now in this section um he's going to be the man to answer um but also I'm going to put a lot of secondary readings up again in the in the document um so you'll be able to find them all let's suppose switch gears slightly I want to keep the focus on feelings although what I'm going to be talking about here are not what might be called like full-blooded emotion States um but I want to talk talk about something a little bit more subtle so one thing that might seem really really obvious kind of so obvious that we often Overlook it when things are going okay is that it generally feels like something to be an agent so when we're going about whatever business we have in we always feel like something and it's not entirely clear how best to characterize that so I want to draw on some work in phenomenology that some active inference theorists have also drawn on and made use of because I think this is a really good grounding for some of the things I'm going to go on to say um certainly proved to be a good grounding for some of the work that I'm going to highlight so these are two books by uh Matthew Ratcliffe who I think is at the University of York and has done really really cool work in kind of really getting to grips with the what it's likeness to live in certain

conditions and Radcliffe has he argues that he's identified this certain area uh this space of feelings that have been largely neglected in The Wider literature and he describes these as existential feelings um so he says existential feelings are uh quote they are not intentional States directed at however many objects and they are not feelings of the body or some part of it instead they amount to a felt sense of belonging in the world so the main difference between a full-blooded emotion State and an existential feeling is that the emotion state is about something okay so if I'm uh disgusted I'm typically well all the time I'm disgusted about something in the world okay so emotions are intentional in that Radcliffe elaborates on existential Feelings by um I'm going to quote this at length because I quite like it he says people sometimes talk of feeling alive dead distance detached dislodged estranged isolated otherworldly indifferent to everything overwhelmed suffocated cut off lost disconnected out of sorts not oneself out of touch with things out of it not quite with it separate in harmony with things at peace with things there are reference to feelings of unreality heightened existence surreality familiarity unfamiliarity strangeness isolation emptiness belonging being at home in the world being at one with things significance insignificance and the list goes on this is uh this really really cool work and I recommend everybody to go and read Matthew Ratcliffe but um some of the work I really like in predictive processing and active inference has taken this idea of an existential feeling and what they've recognized is that these kinds of feelings are really about how we are orientated in regards to the world and the affordances in it so their their background bodily states of action Readiness so how are we relating to um like how do we find ourselves in relation to the world so we're talking about the general structuring that's present in our everyday phenomenology okay so I really want to make that clear that what this work is doing is building a bridge between computational models and the way even the underpinnings of those models in neurobiology and phenomenological experience so the thing that I need to introduce and this can be uh it's fairly straightforward but it can be a little a little bit tricky at times is this notion of aerodynamics okay this is really important so like I just said aerodynamics is going to provide this conceptual bridge between different levels of analysis and it elaborates on how systems are sensitive to the success or failure of their own action policies unfolding across multiple time scales so the recognition here is that we talk a lot about action policies and expected States but the reality is that we have a multitude of goals and action policies and expected states that are interlocking overlapping constantly shifting okay the picture is messy and so I like this quote from the 2019 paper that says the roller coaster of continual increases and decreases of errors that accompany life become expected and are

folded into our expectations for such a system it becomes important not only to track the constantly fluctuating instantaneous errors but also to pay attention to the Dynamics of error reduction over longer time scales so the systems the upshot of this is that these authors believe building on some other work in computational modeling that systems must track the rate of change in overall error reduction relative to the system's expectations so this is um the kind of global upwards or downwards Trend in prediction error that's what the system is sensitive to sensitive to how that relates to the the idea here is that aerodynamics provide a system with a much broader ongoing sense of how they're doing to their multitude of goals over time a big step here is that those changes in the rate of reduction feed back into the system so they're folded back within the system and they serve the role of modulating Precision on action policies so let's say I'm engaged in some tasks I have a particular expectation about how I'm going to do it that task and when I actually engage with the task I do slightly worse than expected the idea there is that that rate of change so the rate of error error reduction is going to decrease until I'm reducing error worse than my expectations and in a typical healthily functioning system that rate of change the sensitivity to that rate of change is going to feed back and reset the Precision waiting on those actions so it's gonna reduce my confidence in that particular action policy such that I might try a different strategy I might try a different tactic or I might you know I might just throw whatever it is that I'm doing away and go and do something else so it's we're starting to see this picture of through expectations sensitivity to change and feedback onto Precision waiting it's starting to dictate how I manage my action policies so there's this ongoing reciprocal relationship between aerodynamics and my future expectations now crucially the claim here is that on the subject level all of this manifests as these existential feelings of some kind so the way that I feel on a kind of day-to-day basis is a reflection of how I'm doing in terms of these sensitivity to aerodynamics so when I fail at the task and precision waiting is down regulated on that particular action policy the thing that actually that kind of oomph that moves me to go and do something else my embodied feelings of frustration that I experience so our as conscious agents those embodied feelings are what are tuning us towards acting more effectively so optimal behavior on this account is characterized as a feeling of a feeling of grip towards a shifting field of affordances so I'm going to unpack this quickly but the feeling of grip in a nutshell is just this feeling that we're engaging with our affordances in the best way that we can and that we're either meeting or exceeding our expectations for prediction error minimization and what they mean by a field of affordances and this is again really important um a landscape of affordances is just the total

opportunities in my niche so my landscape of affordances right now is Brighton okay it's not really going to change very quickly unless I get on a plane and fly somewhere else but affordances aren't neutral in that way they don't just exist there the affordances that stand out to me as relevant or attractive is going to change quite rapidly depending on what it is I'm doing on my internal State on the context and so on so right now my water bottle stands out as attractive a couple of keys on my keyboard stand out as attractive but that's going to shift quite rapidly once this lecture is over and I turn to do something else so the idea of grip is that we are kind of we exhibit a stability in our Attunement such that we can remain attuned as that field changes um and this is part of the skilled intentionality framework which I won't get into because I don't want theoretical overload but I've put some papers up on this slide as references and of course the reading will be in the um but skilled intentionality refers to the selective openness and responsiveness to multiple relevant affordances in a concrete situation so it's just this recognition that things change quickly for multiple reasons and we really need to exhibit a kind of flexibility and stability in those engagements and aerodynamics has been applied in lots of very very cool ways I can't recommend this work highly enough so aerodynamics I won't run through how everything has been explained just for time reasons but um you can kind of take a look at this yourself but aerodynamics has been proposed as a more complete answer to the dark room problem that we were talking about earlier because it can explain the drive for exploration and curiosity because it literally feels good to minimize prediction errors better than expected okay so again that rate of change manifests as embodied feeling so um we're always looking for opportunities to do better than expected okay and kind of this has been suggested to account for feelings of boredom prompting us to go out and find new things to do and similarly aerodynamics has provided a computational account of play which is for a while has remained a kind of mystery because it's on the face of it like why would we engage in play when it's metabolically costly and we need that energy for other things you know like Lisa film and Barrett says like managing energy is super important so why do baby animals burn it all engaging in playful Behavior aerodynamics has a has a story for us there and there are also aerodynamics based accounts of happiness and well-being and depression as well so I won't go into the details there but it turns out that aerodynamics and this connection to phenomenology is this super malleable flexible um kind of uh feature of the framework that really underpins a lot of what we experience and do as active inference agents so I'm gonna talk a little bit about addiction got some time here things are going well so I take um so so I had some options here because I wanted to use a case study that really

brought out uh some of the things I was talking about and as I've just said aerodynamics has been applied in a bunch of different areas and it's all very much worth talking about but I chose addiction because uh well as you'll see as I as I kind of make clear I think it also brings out not just the way that all of this fits together but also why it's good and why seeing people as active inference agents and broader structures as active inference structures is actually a worthwhile thing to do so addiction was once viewed as a moral failing um so the kind of Victorian addict is too rotten to make the right choice to self-indulgent I say Victorian unless you're Peter Hitchens and then you still subscribe to this view but that's the old-fashioned view okay there was something wrong with the addict some moral or spiritual feeling and gradually that changed to what the dominant view is today which is the disease model um and the idea here is that addiction is a disease and that disease is characterized by changes in the brain which changes to brain structures which mean the addict um can't help but act in a certain way when confronted with a certain stimulus so it's this loss of control that's engendered through these brain structures and this is this is the idea of it being a disease the work that I'm going to describe now um I won't go into too much detail here but it's it's it's kind of founded on this thought that this disease model can't quite it can't be quite right because these changes in brain structure essentially they like that's what the brain does like we it's just learning like this is what we do all the time like essentially me completing my PhD is a process of me undergoing changes to my brain um such that I'm kind of uh disposed to act in certain ways um so there's no so Mark Lewis I think it's Mark Lewis in his book 2018 the biology of Desire um argues look addiction is more of just a dysfunction in learning it's not really a disease so we need to think about it um a little bit differently and this work by Miller and colleagues 2020 um that's where it picks up okay so the idea is that those brain changes engendered through exposure to a stimulus in this case we're going to talk about a drug okay we're going to leave behavioral addiction aside for a second um that clearly plays a critical role okay so nobody is going to deny the kind of chemical like the role the importance of the biology here um that's that's going on in the brain and of course different drugs have different chemical profiles in that regard um but on this view we're going to kind of shift Focus from that slightly and we're going to have a much more active picture where we view addiction as a self-organizing process of whole agent environment systems Okay so um I have this extract here it is in the dynamic interaction between the agent and its environment that addiction is born and endures okay so the harm of addiction emerges from a breakdown in the broader stable dynamics that typically underpin a flourishing life work family friends health and so on so these you know

typically flourishing life has a broad kind of bush of concerns over different time scales in different areas and what we're going to see is that the harm of addiction is in a narrowing of those concerns so what happens is when somebody takes a drug um to some people who are exposed to drugs that drug is going to through direct action on some neural mechanisms is going to signal to the agent that they've done better than expected at reducing prediction error of course this is illusory they haven't really done better than expected in anything that matters that the signal they get is wow like we have just reduced prediction error at this insane rate um and repeated use through this aerodynamics mechanism repeated use is going to tune our expectations about the kinds of prediction errors slopes the kind of slopes of prediction error that are available to us in the environment and we're going to start to expect those vertiginous slopes of error reduction if we continue to engage with the drug and once those expectations become deeply ingrained um anything else we do is going to start to feel like failure okay so if I get used to taking cocaine all the time and then I have to sit through a family meal where I don't have any drugs and I have to kind of sit still and there's no dopamine available I don't have my phone that's I'm going to experience that as feelings of frustration and as if I'm being prevented from doing something I want to do and I'm going to experience that as doing worse than expected okay so I've learned these expectations for prediction error minimization and I'm not getting them unless I engage with the drug and of course it's worth noting that in many cases these drugs are going to have a chemical withdrawal effects that compound that as well especially things like nicotine and alcohol in particular is very nasty so alcohol can kill you if you uh if you try and withdraw from it too quickly so they really do pull the agent into grip so what happens here is the agent develops a kind of sub-optimal grip so you start to enact a new world a new identity so instead of doing all the things you once cared about like different Hobbies you start to hang out with different people you start to frequent different places like a dealer's House or wherever and very slowly you start to carve out essentially a niche the addict enacts a particular Niche through their actions and they have quite a good grip on that Niche it's just that it doesn't speak to a broader array of concerns what's really important is that this is not this pushes back against a disease model because on this view the active inference agent the well the system is doing essentially what it was evolved to do so um there's another quote from the paper here that I like that says their habits allow them to remain well attuned and keep prediction errors under tight control so long as they remain within the narrow confines of such a niche so it's not the and this is important because it shows that allostatic control and prediction error minimization are not the same thing they come apart and this really it speaks

to a question that was asked after Abel's session recently about what this prediction error actually look like well in this case you you have a system that has learned the engagement with certain affordances gives it it's an illusion but it has this feedback that says this is you know this is the state you expect to occupy oh this is a state that has the agent has learned to expect to occupy and it's carved out a niche of affordances that means it's very likely to occupy that state so it actually has a really good grip on the situation it's just that um that Niche is very narrow and it's constricted meaning um it's simply that kind of uh those habits and that web of Habitual action policies are just not well suited um to maintained good allostatic control so everything else in the Alex Life Starts to suffer so we can kind of think of this as a form of Niche construction gone wrong and I think we're going to look at Niche Construction in the in the coming weeks as well so why is this whole agent well so I should say just to back up a second so this is what I like about this account of addiction is that it's a thoroughly an active account of addiction it really brings out the kind of inactive core of active inference so the addict is addiction becomes an identity it becomes a world that the person actually produces through their actions so rather than the disease model where you have a person and a stimulus and exposure you know whenever they're exposed to that stimulus they're going to have problems because of these brain changes what this does Justice do is actually the activity of the agent so they're not accidentally bumping into this stimulus and the environment they're Del and they're not just seeking it out but they're deliberately restructuring their environment to make this state more and more likely over time so it really does does Justice to the efficacy and the agency of the agent rather than giving that agency a more backseat role that it takes in the disease model but the other thing that it does because it's this inactive uh picture is that it also does Justice to the environmental features so it also um it kind of speaks to the fact that addiction is not just all in the head actually the real um some of the problems of Addiction come with the fact that it is this um procedural constructive process and one of my favorite experiments of all time is um I don't know if people are familiar with the Rat City experiments this is why I've got the image up here and this is a very cool experiment where originally it was found that if you exposed rats to a drug-laced stimulus and environment I think it was um I think it was heroin laced water or it might have been cocaine um but the original experiments found that like nearly all the rats would kill themselves with the drug so they would just keep keep keep at the drug until um until they died and it was a very very very high percentage of the rats would engage in that destructive behavior until the experiments were re-run um sometime later and instead of having the wrap just in a boring

cage on its own it had the rats in Rat Park which was a very social exciting brightly colored environment with lots of different affordances and what they found was that given the same Choice A vastly reduced proportion of the rats would choose to engage with the drug okay so it's capturing the importance of Niche construction that I think is really nice here and it also speaks to the role played by embodied affect in phenomenology there's something that it's like to be an addict and that really matters okay it really matters that the addict has these embodied experiences when they're engaging with particular sets of affordances so that's really summed up in the overview that I've put here so these were the things that I wanted to emphasize um active inference agents are embodied they're embedded in a landscape that landscape is constituted by particular affordances and we can distinguish between a landscape of affordances and a field of affordances and that distinction really speaks to the inherent affectivity and normativity of affordances okay this which affordances stand out to a certain particular time which exert a pull over us um an active inference agents enact their worlds through action and this is nowhere starker than in the case of addiction active inference agents are effective so emotion plays this super crucial role keeping is keeping us tuned keeping us in that uh grip with our affordances and helping us contextualize those important physiological changes um active inference agents are agents okay so we do have a way to describe Behavior as agential and motivated and we do have these cognitive architectures that can make sense of agential action and active inference agents have this inbuilt sense of self that comes with agency as well so that's pretty much it this final slide just goes over some of the things that we can expect so this what I've done today I hope has laid a good groundwork and has installed some higher level expectations to learn more about active inference and what we're going to do in the subsequent weeks is now scale this up to see how when you have more than one active inference agent together how the expectations of some can impact the expectations of others and individuals and I look forward to to learn more about that but as for me uh that's it Amazing Ben awesome work what can people look forward to next week or in two weeks if they join the discussions section like what's it going to be like or who would you encourage to join um I would definitely I mean the more people that come along I think the better it's going to be I'm really looking forward to it I'm I'm kind of just in the first place Intrigued to see that we we covered so much ground I'm just interested to see where people want to go with this and see where people's interests lie um I assume that the people are going to have questions and I think maybe the best way to do it would be to explore some of people's questions during the discussion we can use them to kick things off and keep things going and but I think it's

likely that that disco we we have a whole hour so we've certainly got time to really dig into things in some detail and like I said before we're going to have Mark Miller with us and Mark is just awesome in discussion it's very very passionate and enthusiastic about his work and he can do a much better job than I can in explaining it so I just um just to have the opportunity to speak with Mark about his work I would I would say definitely come along all right anything else otherwise again awesome job uh no I think that's it cool thanks for listening I guess well thanks it was a pleasure it was ambitious to to what you did elegantly with no equations or getting snagged on any of the potentially infinite philosophical and uh tactical and normative and all these different layers of Challenge and diversions so to really um to really cut a path there is great yeah I think it just speaks to there's there's so much good philosophical work in these areas in active influence and predictive processing and I should say again I mean I I we didn't really do justice to any of the work I talked about so that's the final thing I would say is definitely you know the the document is there with the reading and please go and um actually look at the work of these people because it's it's something that's really really good till next time cool thank you very much